

INDUSTRIAL TRAINING PROJECT

Smart Paper Mill Production Monitoring Dashboard

A real-time, role-based web dashboard for monitoring paper production
Industrial training carried out at SATIA INDUSTRIES LIMITED

Submitted by

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Problem & Objectives

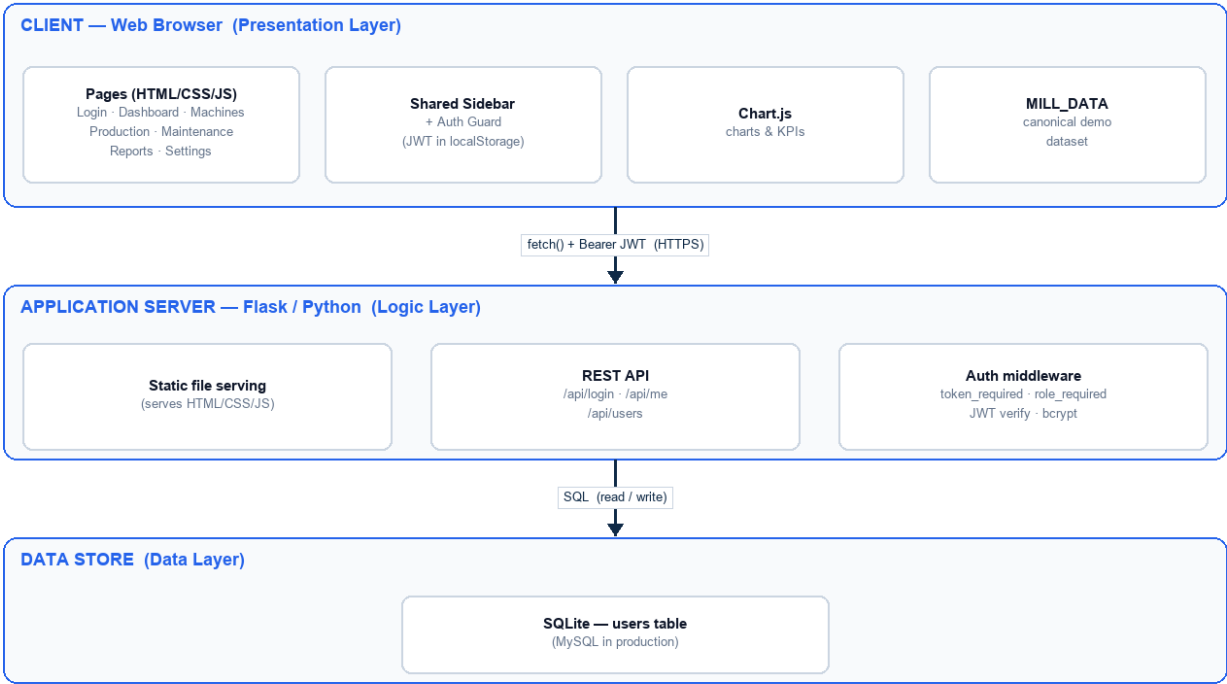
- **Problem:** production, quality (GSM / moisture), machine status and downtime are recorded **manually** in registers and spreadsheets - no single, real-time view, and problems are caught late.
- **Objective:** a **centralised, role-based web dashboard** for real-time production monitoring.
- Capture output plus **quality and downtime** in a database, with **live machine telemetry** and **automatic threshold alerts**.
- Compute **OEE and KPIs**, export **reports (CSV / PDF)**, and enforce **secure, role-based access** - all on standard student hardware.

Technology Stack

Layer	Technology	Why
Front-end	HTML5, CSS3, JavaScript, Chart.js	Lightweight, no build step
Back-end	Python - Flask (REST API + static)	Same-origin API & UI
Database	SQLite (MySQL is production target)	Zero-config persistence
Security	bcrypt, PyJWT	Hashed passwords + JWT auth
Real-time	Server-Sent Events (SSE)	Live push, zero extra deps
Testing / Deploy	pytest (94 tests); gunicorn / Render	Verified; free hosting

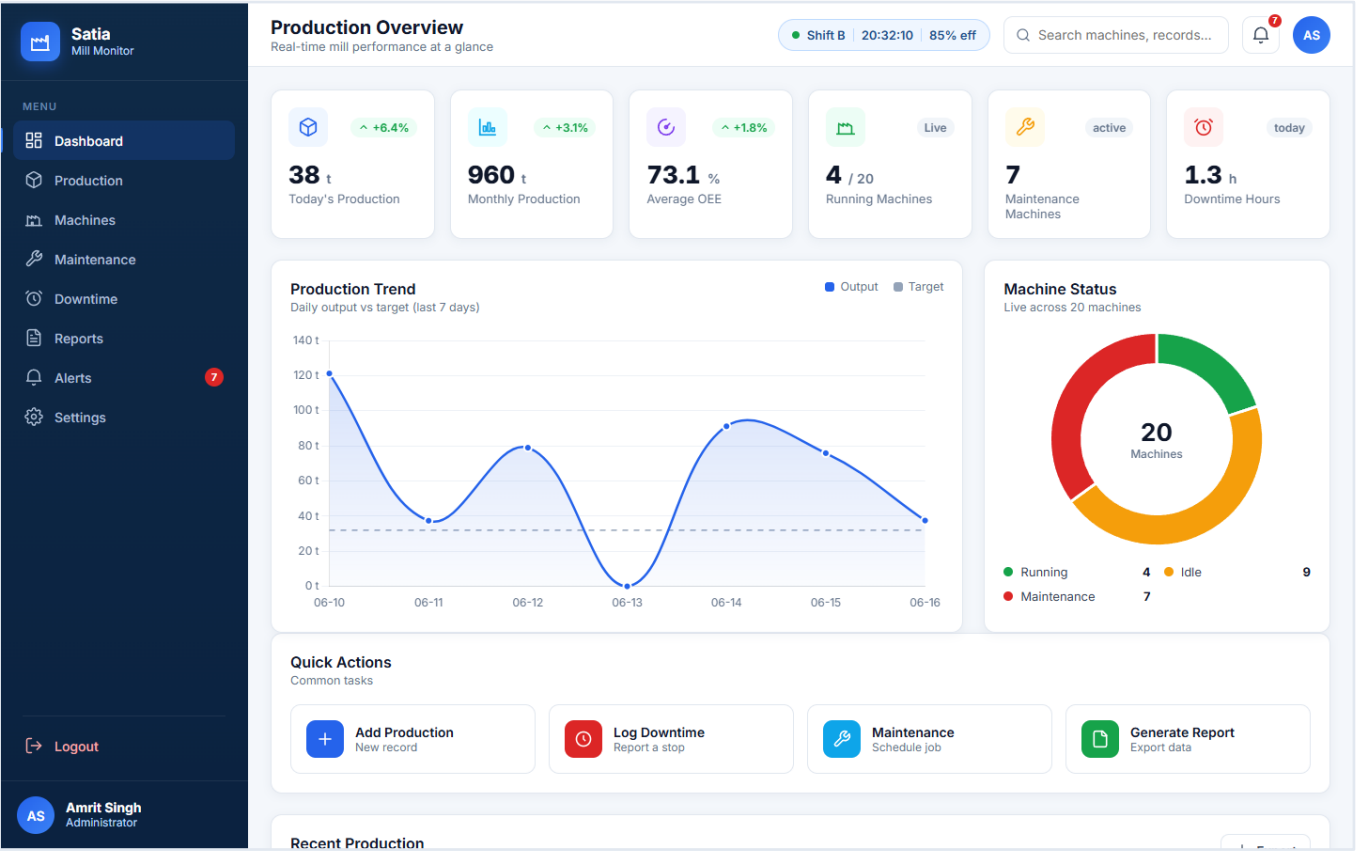
System Architecture

System Architecture — Smart Paper Mill Production Monitoring Dashboard



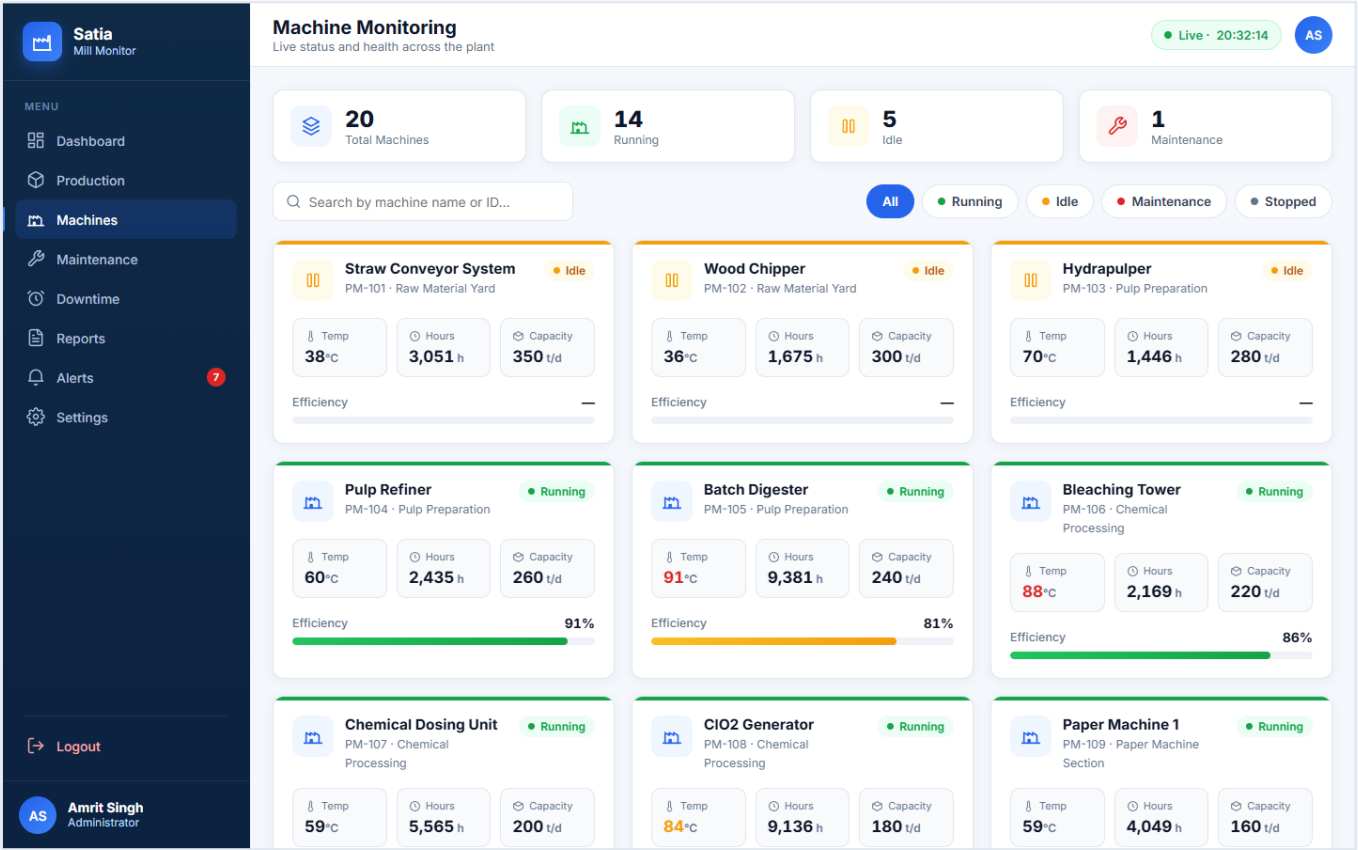
Three-tier: browser front-end -> Flask REST API + SSE -> SQLite database (same origin).

Dashboard



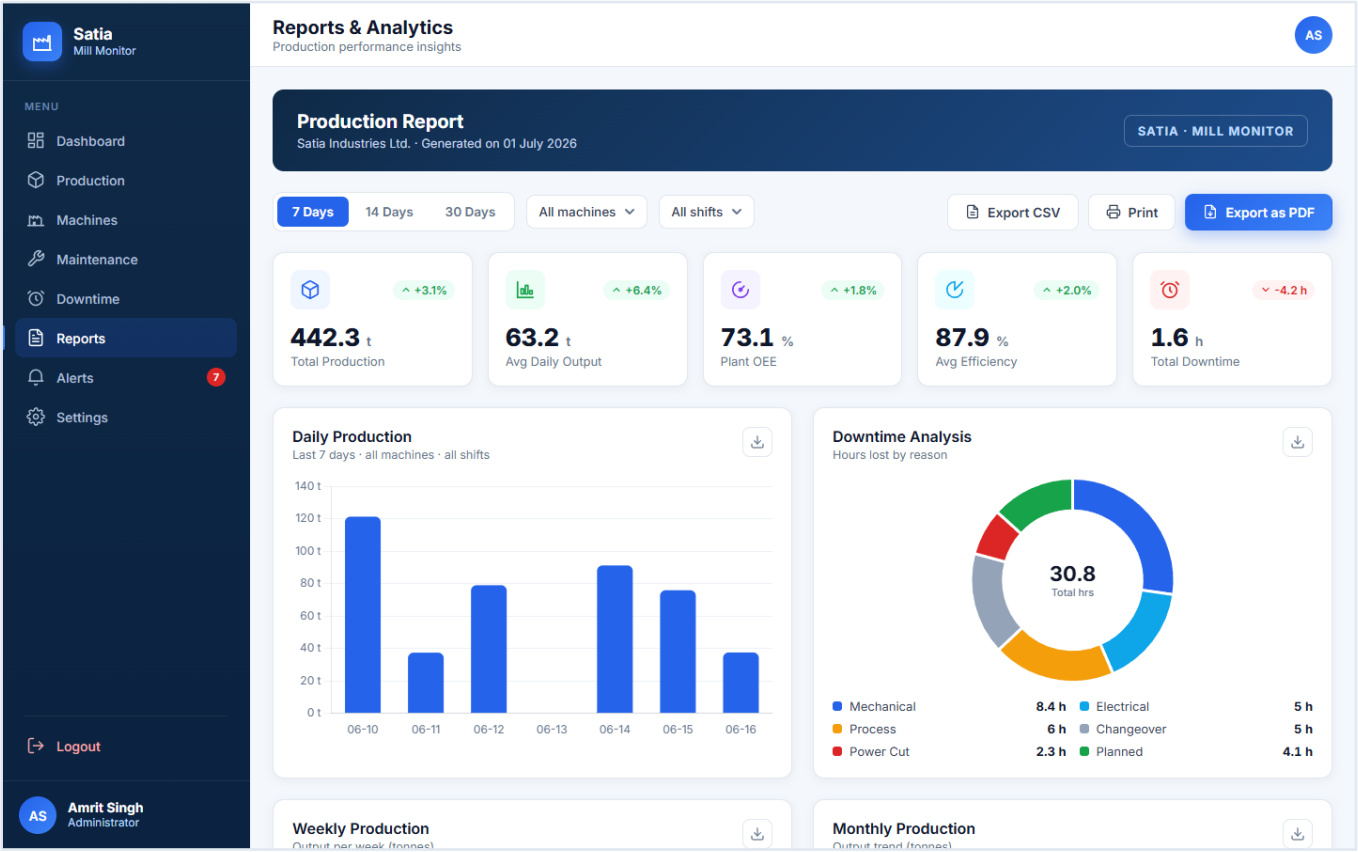
Live overview: KPI cards (incl. average OEE), production-trend and machine-status charts.

Machine Monitoring



Colour-coded status board - Running / Idle / Stopped / Maintenance - with live telemetry.

Reports & Analytics



Filtered reports, plant OEE, downtime breakdown, and CSV / PDF export.

Real-Time, OEE & Alerts

- **Live telemetry:** the server pushes temperature & efficiency to open pages via **SSE**, updating every few seconds without a reload.
- **API-driven:** dashboard KPIs, the daily report and alerts reflect records entered through the app, with graceful fallback when offline.
- **Quality & OEE:** every record captures GSM / moisture / speed against admin-editable limits; **OEE = Availability x Performance x Quality** is charted per machine.
- **Threshold alerts:** out-of-spec quality and over-limit downtime raise alerts that can be filtered by severity and acknowledged.

Testing & Outcomes

- **94 / 94 automated tests passing** (pytest) - authentication, role-based access, full CRUD, quality & thresholds, live stream and audit log.
- A **working, end-to-end application** with real auth, role-based access and **persistence that survives restarts**.
- A complete **documentation set** - SRS, project report and role-based user manuals - and a **deploy-ready** build for free hosting (Render).
- Hands-on experience across **full-stack development, REST APIs, security, testing and deployment**.

Thank You

Questions & Discussion

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Industrial Training Project - Satia Industries Limited